

FreightQuote AI

Master Data Collections — MongoDB

Formatted technical reference with wrapped tables, readable JSON examples, collection fields, and MongoDB indexes.

FreightQuote AI — Master Data Collections (MongoDB)

All collections below are **admin-managed** (created/edited only from the Admin panel), and referenced by the quotation engine.

1. `countries`

FieldType	Notes	
`_id`	ObjectId	
`countryCode`	String	ISO 3166-1 alpha-2, unique
`countryName`	String	
`iso3`	String	alpha-3
`region`	String	Asia, Europe, NAM, LATAM, MEA
`currencyCode`	String	ref `currencies.code`
`dialCode`	String	
`customsUnion`	String	EU / GCC / NAFTA / null
`isActive`	Boolean	
`createdBy`	ObjectId	ref `users`
`createdAt/updatedAt`	Date	

json

```
[
  { "countryCode": "IN", "countryName": "India", "iso3": "IND", "region": "Asia", "currencyCode": "INR", "dialCode": "+91", "
  customsUnion": null, "isActive": true },
  { "countryCode": "AE", "countryName": "United Arab Emirates", "iso3": "ARE", "region": "MEA", "currencyCode": "AED", "dialCode": "+971", "
  customsUnion": "GCC", "isActive": true },
  { "countryCode": "SG", "countryName": "Singapore", "iso3": "SGP", "region": "Asia", "currencyCode": "SGD", "dialCode": "+65", "
  customsUnion": null, "isActive": true },
  { "countryCode": "NL", "countryName": "Netherlands", "iso3": "NLD", "region": "Europe", "currencyCode": "EUR", "dialCode": "+31", "
  customsUnion": "EU", "isActive": true },
  { "countryCode": "US", "countryName": "United States", "iso3": "USA", "region": "NAM", "currencyCode": "USD", "dialCode": "+1", "
  customsUnion": "USMCA", "isActive": true }
]
```

2. `ports`

FieldType	Notes	
`_id`	ObjectId	
`unlocode`	String	UN/LOCODE, unique (e.g. INMAA)
`portName`	String	
`portType`	String	SEAPORT / AIRPORT / ICD / RAIL
`iataCode`	String	for airports
`city`	String	
`countryCode`	String	ref `countries`
`location`	GeoJSON Point	`{ type:"Point", coordinates:[lng,lat] }`
`timezone`	String	

FieldTypeNotes		
`maxVesselDraftM`	Number	seaport only
`terminals`	Array<String>	
`customsOffice`	String	
`avgDwellTimeHrs`	Number	used by Route agent
`congestionIndex`	Number	0–1, refreshed by agent
`isActive`	Boolean	

json

```
[
  {
    "unlocode": "INMAA", "portName": "Chennai Port", "portType": "SEAPORT", "city": "Chennai", "countryCode": "IN", "location": { "type": "Point", "coordinates": [80.2924, 13.0975] }, "timezone": "Asia/Kolkata", "maxVesselDraftM": 16.5, "terminals": ["CCTL", "CITPL"], "customsOffice": "INMAA1", "avgDwellTimeHrs": 62, "congestionIndex": 0.42, "isActive": true },
    {
      "unlocode": "INNSA", "portName": "Jawaharlal Nehru Port (Nhava Sheva)", "portType": "SEAPORT", "city": "Navi Mumbai", "countryCode": "IN", "location": { "type": "Point", "coordinates": [72.9490, 18.9490] }, "timezone": "Asia/Kolkata", "maxVesselDraftM": 15.0, "terminals": ["NSICT", "BMCT", "GTI"], "customsOffice": "INNSA1", "avgDwellTimeHrs": 48, "congestionIndex": 0.55, "isActive": true },
      {
        "unlocode": "AEJEA", "portName": "Jebel Ali", "portType": "SEAPORT", "city": "Dubai", "countryCode": "AE", "location": { "type": "Point", "coordinates": [55.0272, 25.0110] }, "timezone": "Asia/Dubai", "maxVesselDraftM": 17.0, "terminals": ["T1", "T2", "T3"], "customsOffice": "AEJEA1", "avgDwellTimeHrs": 30, "congestionIndex": 0.21, "isActive": true },
        {
          "unlocode": "SGSIN", "portName": "Port of Singapore", "portType": "SEAPORT", "city": "Singapore", "countryCode": "SG", "location": { "type": "Point", "coordinates": [103.8198, 1.2644] }, "timezone": "Asia/Singapore", "maxVesselDraftM": 18.0, "terminals": ["Tuas", "Pasir Panjang"], "customsOffice": "SGSIN1", "avgDwellTimeHrs": 24, "congestionIndex": 0.18, "isActive": true },
          {
            "unlocode": "NLRTM", "portName": "Port of Rotterdam", "portType": "SEAPORT", "city": "Rotterdam", "countryCode": "NL", "location": { "type": "Point", "coordinates": [4.1420, 51.9490] }, "timezone": "Europe/Amsterdam", "maxVesselDraftM": 24.0, "terminals": ["Maasvlakte II", "Euromax"], "customsOffice": "NLRTM1", "avgDwellTimeHrs": 36, "congestionIndex": 0.33, "isActive": true },
            {
              "unlocode": "INMAA", "portName": "Chennai International Airport", "portType": "AIRPORT", "iataCode": "MAA", "city": "Chennai", "countryCode": "IN", "location": { "type": "Point", "coordinates": [80.1709, 12.9941] }, "timezone": "Asia/Kolkata", "terminals": ["Cargo Terminal 1"], "customsOffice": "INMAA4", "avgDwellTimeHrs": 8, "congestionIndex": 0.15, "isActive": true },
              {
                "unlocode": "AEDXB", "portName": "Dubai International Airport", "portType": "AIRPORT", "iataCode": "DXB", "city": "Dubai", "countryCode": "AE", "location": { "type": "Point", "coordinates": [55.3644, 25.2532] }, "timezone": "Asia/Dubai", "terminals": ["Cargo Mega Terminal"], "customsOffice": "AEDXB4", "avgDwellTimeHrs": 6, "congestionIndex": 0.12, "isActive": true }
            ]
          }
        ]
      }
```

3. `tradeLanes`

FieldTypeNotes		
`laneCode`	String	unique, `INNSA-AEJEA-OCEAN`
`originPort`	ObjectId	ref `ports`
`destPort`	ObjectId	ref `ports`
`mode`	String	OCEAN / AIR / GROUND / EXPRESS
`distanceNm`	Number	nautical miles / km
`baseTransitDays`	Number	
`transhipmentPorts`	Array<String>	UN/LOCODEs
`canalsCrossed`	Array<String>	SUEZ, PANAMA
`riskZones`	Array<String>	e.g. RED_ SEA
`isActive`	Boolean	

json

```
[
  {
    "laneCode": "INNSA-AEJEA-OCEAN", "mode": "OCEAN", "distanceNm": 1080, "baseTransitDays": 5, "transhipmentPorts": [], "canalsCrossed": [], "riskZones": [], "isActive": true },
    {
      "laneCode": "INMAA-SGSIN-OCEAN", "mode": "OCEAN", "distanceNm": 1560, "baseTransitDays": 7, "transhipmentPorts": ["MYPKG"], "canalsCrossed": [], "riskZones": [], "isActive": true },
      {
        "laneCode": "INNSA-NLRTM-OCEAN", "mode": "OCEAN", "distanceNm": 6350, "baseTransitDays": 22, "transhipmentPorts": ["AEJEA"], "canalsCrossed": ["SUEZ"], "riskZones": ["RED_SEA"], "isActive": true },
        {
          "laneCode": "MAA-DXB-AIR", "mode": "AIR", "distanceNm": 1670, "baseTransitDays": 2, "transhipmentPorts": [], "canalsCrossed": [], "riskZones": [], "isActive": true }
    ]
  }
```

4. `carriers`

FieldTypeNotes		
`carrierCode`	String	SCAC / IATA, unique
`carrierName`	String	
`mode`	String	OCEAN / AIR / GROUND
`serviceTypes`	Array<String>	ref `serviceTypes.code`
`reliabilityScore`	Number	0–100
`contractTier`	String	SPOT / CONTRACT / NVOCC
`apiEnabled`	Boolean	rate API available
`contactEmail`	String	
`isActive`	Boolean	

json

```
[
  {
    "carrierCode": "MAEU",
    "carrierName": "Maersk Line",
    "mode": "OCEAN",
    "serviceTypes": ["FCL", "LCL"],
    "reliabilityScore": 88,
    "contractTier": "CONTRACT",
    "apiEnabled": true,
    "contactEmail": "booking@maersk.example",
    "isActive": true
  },
  {
    "carrierCode": "MSCU",
    "carrierName": "MSC",
    "mode": "OCEAN",
    "serviceTypes": ["FCL", "LCL", "REEFER"],
    "reliabilityScore": 82,
    "contractTier": "CONTRACT",
    "apiEnabled": true,
    "contactEmail": "ops@msc.example",
    "isActive": true
  },
  {
    "carrierCode": "CMDU",
    "carrierName": "CMA CGM",
    "mode": "OCEAN",
    "serviceTypes": ["FCL", "REEFER"],
    "reliabilityScore": 79,
    "contractTier": "SPOT",
    "apiEnabled": false,
    "contactEmail": "quotes@cmacgm.example",
    "isActive": true
  },
  {
    "carrierCode": "EK",
    "carrierName": "Emirates SkyCargo",
    "mode": "AIR",
    "serviceTypes": ["AIR_GEN", "AIR_EXPRESS"],
    "reliabilityScore": 91,
    "contractTier": "CONTRACT",
    "apiEnabled": true,
    "contactEmail": "cargo@ek.example",
    "isActive": true
  },
  {
    "carrierCode": "AI",
    "carrierName": "Air India Cargo",
    "mode": "AIR",
    "serviceTypes": ["AIR_GEN"],
    "reliabilityScore": 74,
    "contractTier": "SPOT",
    "apiEnabled": false,
    "contactEmail": "cargo@ai.example",
    "isActive": true
  }
]
```

5. `serviceTypes`

FieldTypeNotes		
`code`	String	unique
`label`	String	shown in /ship form
`mode`	String	
`volumetricDivisor`	Number	6000 air, 1000 ocean LCL
`minChargeableKg`	Number	
`defaultTransitDays`	Number	
`sortOrder`	Number	
`isActive`	Boolean	

json

```
[
  {
    "code": "FCL",
    "label": "Ocean Freight - FCL",
    "mode": "OCEAN",
    "volumetricDivisor": 1000,
    "minChargeableKg": 0,
    "defaultTransitDays": 22,
    "sortOrder": 1,
    "isActive": true
  },
  {
    "code": "LCL",
    "label": "Ocean Freight - LCL",
    "mode": "OCEAN",
    "volumetricDivisor": 1000,
    "minChargeableKg": 100,
    "defaultTransitDays": 26,
    "sortOrder": 2,
    "isActive": true
  },
  {
    "code": "AIR_GEN",
    "label": "Air Freight",
    "mode": "AIR",
    "volumetricDivisor": 6000,
    "minChargeableKg": 45,
    "defaultTransitDays": 5,
    "sortOrder": 3,
    "isActive": true
  },
  {
    "code": "AIR_EXPRESS",
    "label": "Express Air",
    "mode": "AIR",
    "volumetricDivisor": 5000,
    "minChargeableKg": 1,
    "defaultTransitDays": 2,
    "sortOrder": 4,
    "isActive": true
  },
  {
    "code": "GROUND",
    "label": "Ground & Rail",
    "mode": "GROUND",
    "volumetricDivisor": 4000,
    "minChargeableKg": 50,
    "defaultTransitDays": 6,
    "sortOrder": 5,
    "isActive": true
  }
]
```

6. `containerTypes`

FieldTypeNotes		
`code`	String	20GP, 40HC... unique
`description`	String	

FieldTypeNotes		
`teu`	Number	
`internalLengthM/WidthM/HeightM`	Number	
`maxPayloadKg`	Number	
`capacityCbm`	Number	
`isReefer`	Boolean	
`isActive`	Boolean	

json

```
[
  { "code": "20GP", "description": "20ft General Purpose", "teu": 1, "internalLengthM": 5.90, "internalWidthM": 2.35, "internalHeightM": 2.39, "maxPayloadKg": 28200, "capacityCbm": 33.2, "isReefer": false, "isActive": true },
  { "code": "40GP", "description": "40ft General Purpose", "teu": 2, "internalLengthM": 12.03, "internalWidthM": 2.35, "internalHeightM": 2.39, "maxPayloadKg": 26680, "capacityCbm": 67.7, "isReefer": false, "isActive": true },
  { "code": "40HC", "description": "40ft High Cube", "teu": 2, "internalLengthM": 12.03, "internalWidthM": 2.35, "internalHeightM": 2.69, "maxPayloadKg": 26460, "capacityCbm": 76.3, "isReefer": false, "isActive": true },
  { "code": "20RF", "description": "20ft Reefer", "teu": 1, "internalLengthM": 5.44, "internalWidthM": 2.29, "internalHeightM": 2.27, "maxPayloadKg": 27400, "capacityCbm": 28.3, "isReefer": true, "isActive": true },
  { "code": "45HC", "description": "45ft High Cube", "teu": 2.25, "internalLengthM": 13.56, "internalWidthM": 2.35, "internalHeightM": 2.69, "maxPayloadKg": 27700, "capacityCbm": 86.0, "isReefer": false, "isActive": true }
]
```

7. `cargoTypes`

FieldTypeNotes		
`code`	String	unique
`label`	String	
`isHazardous`	Boolean	
`imoClass`	String	1–9, null if not DG
`requiresTempControl`	Boolean	
`tempRangeC`	Object	`{min,max}`
`handlingSurchargePct`	Number	applied by Pricing agent
`restrictedCountries`	Array<String>	
`isActive`	Boolean	

json

```
[
  { "code": "GEN", "label": "General Cargo", "isHazardous": false, "imoClass": null, "requiresTempControl": false, "handlingSurchargePct": 0, "restrictedCountries": [], "isActive": true },
  { "code": "PERISH", "label": "Perishable / Food", "isHazardous": false, "imoClass": null, "requiresTempControl": true, "tempRangeC": { "min": 2, "max": 8 }, "handlingSurchargePct": 12, "restrictedCountries": [], "isActive": true },
  { "code": "PHARMA", "label": "Pharmaceuticals", "isHazardous": false, "imoClass": null, "requiresTempControl": true, "tempRangeC": { "min": -20, "max": 8 }, "handlingSurchargePct": 18, "restrictedCountries": [], "isActive": true },
  { "code": "DG3", "label": "Flammable Liquids", "isHazardous": true, "imoClass": "3", "requiresTempControl": false, "handlingSurchargePct": 25, "restrictedCountries": ["SG"], "isActive": true },
  { "code": "LIION", "label": "Lithium Batteries", "isHazardous": true, "imoClass": "9", "requiresTempControl": false, "handlingSurchargePct": 30, "restrictedCountries": [], "isActive": true },
  { "code": "AUTO", "label": "Automotive Parts", "isHazardous": false, "imoClass": null, "requiresTempControl": false, "handlingSurchargePct": 5, "restrictedCountries": [], "isActive": true }
]
```

8. `commodities` (HS Codes)

FieldTypeNotes		
`hsCode`	String	6–8 digit, unique
`description`	String	
`cargoTypeCode`	String	ref `cargoTypes`

FieldTypeNotes		
`defaultDutyPct`	Number	
`requiresLicense`	Boolean	
`isActive`	Boolean	

json

```
[
  { "hsCode": "620342", "description": "Men's cotton trousers", "cargoTypeCode": "GEN", "defaultDutyPct": 12.0, "requiresLicense": false, "isActive": true },
  { "hsCode": "870899", "description": "Motor vehicle parts, other", "cargoTypeCode": "AUTO", "defaultDutyPct": 10.0, "requiresLicense": false, "isActive": true },
  { "hsCode": "300490", "description": "Medicaments, packaged doses", "cargoTypeCode": "PHARMA", "defaultDutyPct": 5.0, "requiresLicense": true, "isActive": true },
  { "hsCode": "850760", "description": "Lithium-ion accumulators", "cargoTypeCode": "LIION", "defaultDutyPct": 15.0, "requiresLicense": true, "isActive": true },
  { "hsCode": "080450", "description": "Mangoes, fresh or dried", "cargoTypeCode": "PERISH", "defaultDutyPct": 30.0, "requiresLicense": false, "isActive": true }
]
```

9. `packagingTypes`

FieldTypeNotes		
`code`	String	unique
`label`	String	
`defaultTareKg`	Number	
`stackable`	Boolean	
`isActive`	Boolean	

json

```
[
  { "code": "PLT_EUR", "label": "Euro Pallet (1200x800)", "defaultTareKg": 25, "stackable": true, "isActive": true },
  { "code": "PLT_STD", "label": "Standard Pallet (1200x1000)", "defaultTareKg": 30, "stackable": true, "isActive": true },
  { "code": "CTN", "label": "Carton Box", "defaultTareKg": 0.5, "stackable": true, "isActive": true },
  { "code": "CRATE", "label": "Wooden Crate", "defaultTareKg": 40, "stackable": false, "isActive": true },
  { "code": "DRUM", "label": "Steel Drum (200L)", "defaultTareKg": 18, "stackable": false, "isActive": true }
]
```

10. `incoterms`

FieldTypeNotes		
`code`	String	unique
`label`	String	
`version`	String	2020
`freightPaidBy`	String	SHIPPER / CONSIGNEE
`insurancePaidBy`	String	
`originChargesIncluded`	Boolean	
`destChargesIncluded`	Boolean	
`isActive`	Boolean	

json

```
[
  { "code": "EXW", "label": "Ex Works", "version": "2020", "freightPaidBy": "CONSIGNEE", "insurancePaidBy": "CONSIGNEE", "originChargesIncluded": false, "destChargesIncluded": false, "isActive": true },
  { "code": "FOB", "label": "Free On Board", "version": "2020", "freightPaidBy": "CONSIGNEE", "insurancePaidBy": "CONSIGNEE", "originChargesIncluded": true, "destChargesIncluded": false, "isActive": true },
  { "code": "CIF", "label": "Cost, Insurance and Freight", "version": "2020", "freightPaidBy": "SHIPPER", "insurancePaidBy": "SHIPPER", "originChargesIncluded": true, "destChargesIncluded": false, "isActive": true },
  { "code": "DAP", "label": "Delivered At Place", "version": "2020", "freightPaidBy": "SHIPPER", "insurancePaidBy": "SHIPPER", "originChargesIncluded": true, "destChargesIncluded": true, "isActive": true },
  { "code": "DDP", "label": "Delivered Duty Paid", "version": "2020", "freightPaidBy": "SHIPPER", "insurancePaidBy": "SHIPPER", "originChargesIncluded": true, "destChargesIncluded": true, "isActive": true }
]
```

11. `chargeHeads`

FieldTypeNotes		
`code`	String	unique
`label`	String	
`category`	String	ORIGIN / FREIGHT / DEST / CUSTOMS
`calcBasis`	String	PER_CONTAINER / PER_KG / PER_CBM / PER_SHIPMENT / PERCENT
`defaultValue`	Number	
`currency`	String	
`taxable`	Boolean	
`isActive`	Boolean	

json

```
[
  { "code": "OFR", "label": "Ocean Freight", "category": "FREIGHT", "calcBasis": "PER_CONTAINER", "defaultValue": 950, "currency": "USD", "taxable": false, "isActive": true },
  { "code": "BAF", "label": "Bunker Adjustment Factor", "category": "FREIGHT", "calcBasis": "PER_CONTAINER", "defaultValue": 180, "currency": "USD", "taxable": false, "isActive": true },
  { "code": "THC_O", "label": "Terminal Handling - Origin", "category": "ORIGIN", "calcBasis": "PER_CONTAINER", "defaultValue": 8500, "currency": "INR", "taxable": true, "isActive": true },
  { "code": "THC_D", "label": "Terminal Handling - Destination", "category": "DEST", "calcBasis": "PER_CONTAINER", "defaultValue": 210, "currency": "USD", "taxable": true, "isActive": true },
  { "code": "DOC", "label": "Documentation Fee", "category": "ORIGIN", "calcBasis": "PER_SHIPMENT", "defaultValue": 3500, "currency": "INR", "taxable": true, "isActive": true },
  { "code": "CUSTCL", "label": "Customs Clearance", "category": "CUSTOMS", "calcBasis": "PER_SHIPMENT", "defaultValue": 5000, "currency": "INR", "taxable": true, "isActive": true },
  { "code": "INS", "label": "Cargo Insurance", "category": "FREIGHT", "calcBasis": "PERCENT", "defaultValue": 0.35, "currency": "USD", "taxable": false, "isActive": true }
]
```

12. `currencies` + `exchangeRates`

`currencies`

FieldType	
`code`	String (unique)
`name`	String
`symbol`	String
`decimalPlaces`	Number
`isBaseCurrency`	Boolean
`isActive`	Boolean

json

```
[
  { "code": "USD", "name": "US Dollar", "symbol": "$", "decimalPlaces": 2, "isBaseCurrency": true, "isActive": true },
  { "code": "INR", "name": "Indian Rupee", "symbol": "₹", "decimalPlaces": 2, "isBaseCurrency": false, "isActive": true },
  { "code": "EUR", "name": "Euro", "symbol": "€", "decimalPlaces": 2, "isBaseCurrency": false, "isActive": true },
  { "code": "AED", "name": "UAE Dirham", "symbol": "د.إ", "decimalPlaces": 2, "isBaseCurrency": false, "isActive": true },
  { "code": "SGD", "name": "Singapore Dollar", "symbol": "S$", "decimalPlaces": 2, "isBaseCurrency": false, "isActive": true }
]
```

`exchangeRates`

FieldType	
`baseCurrency`	String
`quoteCurrency`	String
`rate`	Number
`effectiveFrom`	Date
`effectiveTo`	Date
`source`	String

json

```
[
  { "baseCurrency": "USD", "quoteCurrency": "INR", "rate": 87.42, "effectiveFrom": "2026-08-01T00:00:00Z", "effectiveTo": "2026-08-31T23:59:59Z", "source": "ADMIN" },
  { "baseCurrency": "USD", "quoteCurrency": "EUR", "rate": 0.91, "effectiveFrom": "2026-08-01T00:00:00Z", "effectiveTo": "2026-08-31T23:59:59Z", "source": "ADMIN" },
  { "baseCurrency": "USD", "quoteCurrency": "AED", "rate": 3.67, "effectiveFrom": "2026-08-01T00:00:00Z", "effectiveTo": "2026-08-31T23:59:59Z", "source": "ADMIN" },
  { "baseCurrency": "USD", "quoteCurrency": "SGD", "rate": 1.29, "effectiveFrom": "2026-08-01T00:00:00Z", "effectiveTo": "2026-08-31T23:59:59Z", "source": "ADMIN" }
]
```

13. `rateCards` (base tariff master)

FieldTypeNotes		
`rateCardId`	String	unique
`laneCode`	String	ref `tradeLanes`
`carrierCode`	String	ref `carriers`
`serviceTypeCode`	String	ref `serviceTypes`
`containerTypeCode`	String	null for LCL/AIR
`baseRate`	Number	
`currency`	String	
`rateUnit`	String	PER_CONTAINER / PER_KG / PER_CBM
`charges`	Array	`[[chargeCode, value]]`
`validFrom` / `validTo`	Date	
`isActive`	Boolean	

json

```
[
  { "rateCardId": "RC-2026-0001", "laneCode": "INNSA-AEJEA-OCEAN", "carrierCode": "MAEU", "serviceTypeCode": "FCL", "containerTypeCode": "40HC", "baseRate": 1150, "currency": "USD", "rateUnit": "PER_CONTAINER", "charges": [{ "chargeCode": "BAF", "value": 180 }, { "chargeCode": "THC_O", "value": 8500 }], "validFrom": "2026-08-01", "validTo": "2026-10-31", "isActive": true },
  { "rateCardId": "RC-2026-0002", "laneCode": "INNSA-NLRTM-OCEAN", "carrierCode": "MSCU", "serviceTypeCode": "FCL", "containerTypeCode": "20GP", "baseRate": 1780, "currency": "USD", "rateUnit": "PER_CONTAINER", "charges": [{ "chargeCode": "BAF", "value": 240 }, { "chargeCode": "THC_D", "value": 210 }], "validFrom": "2026-08-01", "validTo": "2026-09-30", "isActive": true },
  { "rateCardId": "RC-2026-0003", "laneCode": "INMAA-SGSIN-OCEAN", "carrierCode": "CMDU", "serviceTypeCode": "LCL", "containerTypeCode": null, "baseRate": 48, "currency": "USD", "rateUnit": "PER_CBM", "charges": [{ "chargeCode": "DOC", "value": 3500 }], "validFrom": "2026-07-15", "validTo": "2026-12-31", "isActive": true },
  { "rateCardId": "RC-2026-0004", "laneCode": "MAA-DXB-AIR", "carrierCode": "EK", "serviceTypeCode": "AIR_GEN", "containerTypeCode": null, "baseRate": 2.85, "currency": "USD", "rateUnit": "PER_KG", "charges": [{ "chargeCode": "DOC", "value": 3500 }], "validFrom": "2026-08-01", "validTo": "2026-09-30", "isActive": true }
]
```

14. `surchargeRules`

FieldTypeNotes		
`ruleCode`	String	unique
`label`	String	
`appliesTo`	Object	{mode, laneCode, cargoTypeCode, portCode} (nulls = any)
`triggerCondition`	String	expression consumed by Pricing agent
`valueType`	String	FLAT / PERCENT
`value`	Number	
`currency`	String	
`priority`	Number	
`isActive`	Boolean	

json

```
[
  {
    "ruleCode": "SUR-PEAK",
    "label": "Peak Season Surcharge",
    "appliesTo": {
      "mode": "OCEAN",
      "laneCode": null,
      "cargoTypeCode": null
    },
    "triggerCondition": "month IN [8,9,10]",
    "valueType": "FLAT",
    "value": 300,
    "currency": "USD",
    "priority": 10,
    "isActive": true
  },
  {
    "ruleCode": "SUR-WAR",
    "label": "War Risk / Red Sea Surcharge",
    "appliesTo": {
      "mode": "OCEAN",
      "laneCode": "INNNSA-NLRTM-OCEAN"
    },
    "triggerCondition": "riskZones CONTAINS 'RED_SEA'",
    "valueType": "FLAT",
    "value": 550,
    "currency": "USD",
    "priority": 5,
    "isActive": true
  },
  {
    "ruleCode": "SUR-DG",
    "label": "Dangerous Goods Surcharge",
    "appliesTo": {
      "mode": null,
      "cargoTypeCode": "DG3"
    },
    "triggerCondition": "cargo.isHazardous == true",
    "valueType": "PERCENT",
    "value": 25,
    "currency": null,
    "priority": 3,
    "isActive": true
  },
  {
    "ruleCode": "SUR-CONG",
    "label": "Port Congestion Surcharge",
    "appliesTo": {
      "mode": "OCEAN",
      "portCode": "INNNSA"
    },
    "triggerCondition": "port.congestionIndex > 0.5",
    "valueType": "FLAT",
    "value": 120,
    "currency": "USD",
    "priority": 8,
    "isActive": true
  }
]
```

15. `marginRules`

FieldTypeNotes		
`ruleCode`	String	unique
`customerTier`	String	STANDARD / SILVER / GOLD / ENTERPRISE
`mode`	String	
`laneCode`	String	nullable
`marginType`	String	PERCENT / FLAT
`marginValue`	Number	
`minMarginUsd`	Number	floor used by Margin agent
`maxDiscountPct`	Number	
`isActive`	Boolean	

json

```
[
  {
    "ruleCode": "MR-STD-OCEAN",
    "customerTier": "STANDARD",
    "mode": "OCEAN",
    "laneCode": null,
    "marginType": "PERCENT",
    "marginValue": 18,
    "minMarginUsd": 120,
    "maxDiscountPct": 5,
    "isActive": true
  },
  {
    "ruleCode": "MR-GOLD-OCEAN",
    "customerTier": "GOLD",
    "mode": "OCEAN",
    "laneCode": null,
    "marginType": "PERCENT",
    "marginValue": 11,
    "minMarginUsd": 90,
    "maxDiscountPct": 12,
    "isActive": true
  },
  {
    "ruleCode": "MR-STD-AIR",
    "customerTier": "STANDARD",
    "mode": "AIR",
    "laneCode": null,
    "marginType": "PERCENT",
    "marginValue": 22,
    "minMarginUsd": 60,
    "maxDiscountPct": 6,
    "isActive": true
  },
  {
    "ruleCode": "MR-ENT-ALL",
    "customerTier": "ENTERPRISE",
    "mode": null,
    "laneCode": null,
    "marginType": "PERCENT",
    "marginValue": 8,
    "minMarginUsd": 75,
    "maxDiscountPct": 15,
    "isActive": true
  }
]
```

16. `customsTariffs`

FieldTypeNotes		
`countryCode`	String	destination
`hsCode`	String	ref `commodities`
`dutyPct`	Number	
`vatPct`	Number	
`otherLevies`	Array	`[{name, pct}]`
`requiredDocs`	Array<String>	
`effectiveFrom`	Date	
`isActive`	Boolean	

json

```
[
  {
    "countryCode": "AE", "hsCode": "620342", "dutyPct": 5.0, "vatPct": 5.0, "otherLevies": [], "requiredDocs": ["COMMERCIAL_INVOICE", "PACKING_LIST", "COO"], "effectiveFrom": "2026-01-01", "isActive": true },
    {
      "countryCode": "NL", "hsCode": "870899", "dutyPct": 4.5, "vatPct": 21.0, "otherLevies": [{ "name": "EU Handling", "pct": 0.5 }], "requiredDocs": ["COMMERCIAL_INVOICE", "PACKING_LIST", "EUR1"], "effectiveFrom": "2026-01-01", "isActive": true },
      {
        "countryCode": "IN", "hsCode": "850760", "dutyPct": 15.0, "vatPct": 18.0, "otherLevies": [{ "name": "Social Welfare Surcharge", "pct": 10 }], "requiredDocs": ["COMMERCIAL_INVOICE", "BIS_CERT", "MSDS"], "effectiveFrom": "2026-04-01", "isActive": true },
        {
          "countryCode": "SG", "hsCode": "080450", "dutyPct": 0.0, "vatPct": 9.0, "otherLevies": [], "requiredDocs": ["PHYTOSANITARY", "COMMERCIAL_INVOICE"], "effectiveFrom": "2026-01-01", "isActive": true }
      ]
}
```

17. `documentTypes`

FieldTypeNotes		
`code`	String	unique
`label`	String	
`issuedBy`	String	SHIPPER / CARRIER / AUTHORITY
`mandatoryFor`	Array<String>	mode or cargo codes
`isActive`	Boolean	

json

```
[
  {
    "code": "COMMERCIAL_INVOICE", "label": "Commercial Invoice", "issuedBy": "SHIPPER", "mandatoryFor": ["OCEAN", "AIR", "GROUND"], "isActive": true },
    {
      "code": "PACKING_LIST", "label": "Packing List", "issuedBy": "SHIPPER", "mandatoryFor": ["OCEAN", "AIR", "GROUND"], "isActive": true },
      {
        "code": "BL", "label": "Bill of Lading", "issuedBy": "CARRIER", "mandatoryFor": ["OCEAN"], "isActive": true },
        {
          "code": "AWB", "label": "Air Waybill", "issuedBy": "CARRIER", "mandatoryFor": ["AIR"], "isActive": true },
          {
            "code": "COO", "label": "Certificate of Origin", "issuedBy": "AUTHORITY", "mandatoryFor": [], "isActive": true },
            {
              "code": "MSDS", "label": "Material Safety Data Sheet", "issuedBy": "SHIPPER", "mandatoryFor": ["DG3", "LIION"], "isActive": true },
              {
                "code": "PHYTOSANITARY", "label": "Phytosanitary Certificate", "issuedBy": "AUTHORITY", "mandatoryFor": ["PERISH"], "isActive": true }
            ]
  }
]
```

18. `customerTiers`

FieldTypeNotes		
`code`	String	unique
`label`	String	
`minAnnualVolumeTeu`	Number	
`discountPct`	Number	
`creditDays`	Number	
`slaResponseHrs`	Number	
`isActive`	Boolean	

json

```
[
  { "code": "STANDARD", "label": "Standard", "minAnnualVolumeTeu": 0, "discountPct": 0, "creditDays": 0, "slaResponseHrs": 24, "isActive": true },
  { "code": "SILVER", "label": "Silver", "minAnnualVolumeTeu": 50, "discountPct": 3, "creditDays": 15, "slaResponseHrs": 12, "isActive": true },
  { "code": "GOLD", "label": "Gold", "minAnnualVolumeTeu": 250, "discountPct": 7, "creditDays": 30, "slaResponseHrs": 6, "isActive": true },
  { "code": "ENTERPRISE", "label": "Enterprise", "minAnnualVolumeTeu": 1000, "discountPct": 12, "creditDays": 60, "slaResponseHrs": 2, "isActive": true }
]
```

19. `masterDataAudit` (admin change log)

FieldTypeNotes		
`collectionName`	String	which master collection
`recordId`	ObjectId	
`action`	String	CREATE / UPDATE / DELETE / ACTIVATE
`changedFields`	Object	`{field: {old, new}}`
`performedBy`	ObjectId	ref `users` (admin)
`performedAt`	Date	
`ipAddress`	String	

json

```
[
  { "collectionName": "rateCards", "recordId": "RC-2026-0001", "action": "UPDATE", "changedFields": { "baseRate": { "old": 1090, "new": 1150 } }, "performedBy": "adm_001", "performedAt": "2026-08-04T09:12:00Z", "ipAddress": "10.4.2.18" },
  { "collectionName": "ports", "recordId": "INNSA", "action": "UPDATE", "changedFields": { "congestionIndex": { "old": 0.38, "new": 0.55 } }, "performedBy": "adm_002", "performedAt": "2026-08-06T14:30:00Z", "ipAddress": "10.4.2.21" },
  { "collectionName": "surchargeRules", "recordId": "SUR-WAR", "action": "CREATE", "changedFields": {}, "performedBy": "adm_001", "performedAt": "2026-07-28T11:05:00Z", "ipAddress": "10.4.2.18" }
]
```

Recommended indexes

js

```
db.ports.createIndex({ unlocode: 1, portType: 1 }, { unique: true })
db.ports.createIndex({ location: "2dsphere" })
db.tradeLanes.createIndex({ originPort: 1, destPort: 1, mode: 1 }, { unique: true })
db.rateCards.createIndex({ laneCode: 1, serviceTypeCode: 1, validFrom: -1, validTo: -1 })
db.rateCards.createIndex({ carrierCode: 1, isActive: 1 })
db.exchangeRates.createIndex({ baseCurrency: 1, quoteCurrency: 1, effectiveFrom: -1 })
db.customsTariffs.createIndex({ countryCode: 1, hsCode: 1 }, { unique: true })
```